

Customer Journey Map

Date	03 October 2026
Team ID	SWTID-2026-3758
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Platform: ServiceNow (Service Catalog, Flow Designer)
Based on the project brief on Skillwallet and the ServiceNow documentation for Service Catalog, Flow Designer, Ask for Approval and Create Catalog Task.

Scenario

Requesting, approving, and fulfilling a Standard Laptop through the ServiceNow Service Catalog, with Flow Designer creating the Catalog Task for the Hardware team automatically once the request is approved.

Personas

Employee (Requester)	Manager (Approver)	Hardware Team Member (Fulfiller)
Needs a standard laptop for work. Wants a quick, simple request and clear updates.	Reviews and approves laptop requests. Wants a clear request summary and a fast approve/reject action.	Configures and delivers laptops. Wants tasks to arrive already created, assigned and described.

Steps

Entice	Enter	Engage	Exit	Extend
How does someone become aware of this process?	What do people experience as they begin the process?	In the core moments of the process, what happens?	What do people experience as the process finishes?	What happens after the experience is over?

Interactions

Steps	What does the person typically experience?	What interactions do they have at each step along the way?
Entice	• Employee realises a new or replacement laptop is needed • Opens the ServiceNow Service Catalog and finds the Standard Laptop item	• Browses the catalog categories • Reads the item description
Enter	• Employee fills in the catalog item form and submits the order • Request (REQ) and Requested Item (RITM) are created • Employee gets a confirmation	• Selects the Standard Laptop item • Submits the order form
Engage	• Manager receives the approval request and approves or rejects it • On approval, the flow runs and creates a Catalog Task • Short description is updated and the task is assigned to the Hardware group	• Approval is requested automatically (Ask for Approval) • Flow Designer triggers on the Service Catalog request • Create Catalog Task action runs
Exit	• Hardware team picks up the task, configures the laptop and closes the task • Requested item is marked complete • Employee is notified the laptop is ready	• Hardware team works the Catalog Task from their task list • Task state is updated to closed complete
Extend	• Employee confirms receipt • Team reviews flow execution and request history • Feedback is collected for improvements	• Employee checks request status • Admin opens Flow Context / execution details

Touchpoints

Things, people and places

Entice	Enter	Engage	Exit	Extend
- Things: ServiceNow portal / Service Catalog - People: Employee - Place: Company network, browser	- Things: Standard Laptop catalog item, REQ and RITM records - People: Employee	- Things: Approval, Flow Designer flow, Catalog Task - People: Manager, ServiceNow admin (flow owner)	- Things: Catalog Task, Hardware assignment group - People: Hardware team member, Employee	- Things: Request history, notifications, flow execution log - People: Employee, ServiceNow admin

Goals & Motivations

"Help me..." / "Help me avoid..."

Entice	Enter	Engage	Exit	Extend
- Help me get a laptop without chasing people - Help me find the right catalog item	- Help me place the order quickly and correctly - Help me avoid filling in the same details again	- Help me approve quickly with all the information I need - Help me avoid manual task creation and assignment	- Help me start configuration promptly with clear task details - Help me know when the laptop is ready	- Help me trust the process next time - Help me improve the flow with real feedback

Positive Moments

What feels enjoyable or productive?

Entice	Enter	Engage	Exit	Extend
• A single catalog makes standard items easy to find	• Simple form with a standard item removes guesswork • Instant confirmation of the request	• Task is created automatically right after approval • No manual hand-off between approver and hardware team • Consistent short description on every task	• Hardware team sees the task already assigned to them • Work begins sooner, fewer delays	• Clear audit trail of the whole request • Process is consistent and repeatable

Negative Moments

What feels frustrating or slow?

Entice	Enter	Engage	Exit	Extend
• Employee may not know which catalog item to pick	• Waiting without visibility once the order is submitted	• Flow pauses at Ask for Approval until the manager responds • Approval can be delayed if the manager is unavailable	• If the flow is misconfigured, the task may not be created or assigned correctly	• Flow errors are only visible to admins via flow execution details

Areas of Opportunity

How might we make each step better?

Entice	Enter	Engage	Exit	Extend
• How might we add a clear description and image to the Standard Laptop item?	• How might we show an expected delivery time on the order form?	• How might we send approval reminders to avoid delays? • How might we add escalation if no response?	• How might we add Hardware team notifications when a task is assigned? • How might we add task checklists for configuration steps?	• How might we add a short satisfaction survey? • How might we extend the same flow pattern to other standard catalog items?

References

<https://www.servicenow.com/docs/r/build-workflows/workflow-studio/build-your-first-flow.html>

<https://docs.servicenow.com/bundle/washingtondc-build-workflows/page/administer/flow-designer/reference/ext-related-files.html> (Create Catalog Task action, Ask for Approval action)

Project Design Phase

Data Flow Diagrams & User Stories

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Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Data flow (text form; the diagram is in the Word/PDF version):

Step	From	Process	To / Data store	Data
1	Employee (Requester)	1.0 Submit Standard Laptop request (Service Catalog)	D1: Requested Items	Order, REQ / RITM
2	1.0	2.0 Approve request (Ask for Approval)	Manager (Approver); D2: Approvals	Request, approve / reject decision
3	2.0 (approved)	3.0 Flow Designer: create Catalog Task, set short description, assign Hardware	D3: Catalog Tasks	Catalog Task
4	3.0	4.0 Configure laptop and close task	Hardware Team (Fulfiller)	Task, completion status
5	D1: Requested Items	Track status	Employee (Requester)	Request status

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Employee (Requester)	Service Catalog	USN-1	As an employee, I can open the Service Catalog and find the Standard Laptop item.	I can see the Standard Laptop item in the catalog	High	Sprint-1
Employee (Requester)	Order Placement	USN-2	As an employee, I can submit a Standard Laptop order from the catalog form.	A request (REQ) and requested item (RITM) are created	High	Sprint-1
Manager (Approver)	Approval	USN-3	As a manager, I can approve or reject the laptop request.	Approval record is updated and the flow continues only on approval	High	Sprint-1
ServiceNow Admin	Flow Designer	USN-4	As an admin, I can build a flow that creates a Catalog Task once the request is approved.	Catalog Task is created automatically after approval	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
ServiceNow Admin	Flow Designer	USN-5	As an admin, I can have the flow update the task short description and assign the Hardware group.	Short description is set and the task is assigned to Hardware	High	Sprint-1
ServiceNow Admin	Flow Assignment	USN-6	As an admin, I can attach the flow to the Standard Laptop catalog item.	Ordering the item triggers the flow	High	Sprint-1
Hardware Team Member	Fulfilment	USN-7	As a hardware team member, I can see the assigned Catalog Task and configure the laptop.	Task appears in the Hardware group queue and can be closed	Medium	Sprint-1
Employee (Requester)	Status Tracking	USN-8	As an employee, I can track the status of my laptop request.	Request status is visible until it is closed	Medium	Sprint-1

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

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Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Service Catalog Request	Browse the catalog and select the Standard Laptop item Submit the order form
FR-2	Request Records	A request (REQ) and requested item (RITM) are created on submission Requester receives confirmation
FR-3	Approval	Approval is requested from the manager (Ask for Approval) Flow continues only after the request is approved
FR-4	Automated Catalog Task	Flow Designer creates a Catalog Task after approval Short description of the task is updated automatically
FR-5	Task Assignment	Catalog Task is assigned to the Hardware assignment group Hardware team can view and work the task

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-6	Flow Assignment & Testing	Flow is attached to the Standard Laptop catalog item End-to-end test of order, approval and task creation

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Standard Laptop is ordered with one simple catalog form; no manual task creation is needed.
NFR-2	Security	Access is controlled by ServiceNow roles and ACLs; only approvers approve and only the Hardware group works the tasks.
NFR-3	Reliability	The flow runs the same way for every approved request, avoiding manual errors; flow execution details can be reviewed.
NFR-4	Performance	The Catalog Task is created immediately after approval, so configuration can begin without delay.
NFR-5	Availability	Solution runs on the cloud-hosted ServiceNow instance, accessible through a web browser.
NFR-6	Scalability	The same flow pattern can be reused for other standard

NFR No.	Non-Functional Requirement	Description
		catalog items and more requests.

Project Design Phase-II

Technology Stack (Architecture & Stack)

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Technical Architecture:

Employee (browser) → ServiceNow Service Catalog → Request / Requested Item → Approval (Manager) → Flow Designer → Create Catalog Task → Hardware group queue.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Employees order the Standard Laptop from the catalog	ServiceNow Service Catalog / web browser
2	Application Logic-1	Automation that runs when a request is submitted	ServiceNow Flow Designer (low-code)
3	Application Logic-2	Manager approval of the request	Ask for Approval action
4	Application Logic-3	Task creation, short description update and assignment to Hardware	Create Catalog Task action, Update Record
5	Database	Requests, requested items, approvals and catalog tasks	ServiceNow platform tables (sc_request, sc_req_item, sc_task)
6	Cloud Database	Data hosted on the ServiceNow cloud instance	ServiceNow instance (dev449683)

S.No	Component	Description	Technology
7	File Storage	Attachments stored with the records	ServiceNow attachments
8	External API	Not required for this project	N/A
9	Infrastructure (Server / Cloud)	Application runs on the ServiceNow cloud platform	ServiceNow Developer Instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Low-code platform features: Flow Designer and Service Catalog	ServiceNow Flow Designer
2	Security Implementations	Role-based access, ACLs and approval controls	ServiceNow roles and ACLs
3	Scalable Architecture	Cloud platform; flow reusable for other catalog items	ServiceNow platform
4	Availability	Cloud-hosted instance accessible from a browser	ServiceNow cloud
5	Performance	Automated task creation removes manual delays	Flow Designer automation

References:

<https://www.servicenow.com/docs/r/build-workflows/workflow-studio/build-your-first-flow.html>